A4 Solutions: Geometry and measurement

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Try every problem properly before reading its solution. Your sketch should match the one described here; if it doesn't, redraw it first.

Practice

1.

(a) $180° - 65° = 115°$: angles on a straight line (*nevenhoeken*) along ℓ.

(b) $65°$: vertically opposite (*overstaande hoeken*) to the given angle.

(c) $65°$: corresponding angles (*F-hoeken*) with the given angle, since $\ell \parallel m$.

(d) $180° - 65° = 115°$: it is on a straight line with the angle in (c). (Or: it is the alternate angle (*Z-hoek*) of the angle at P between ℓ to the right and t downwards, which is $115°$.)

2.

(a) $x + 2x + 3x = 180°$, so $6x = 180°$, $x = 30°$. The angles are $30°, 60°, 90°$.

(b) $180° - 47° - 68° = 65°$.

(c) $90°$ and $180° - 90° - 34° = 56°$.

(d) The interior angle at C is $180° - 130° = 50°$. The exterior angle equals the sum of the two opposite interior angles: $130° = 55° + \angle B$, so $\angle B = 75°$. Check: $55° + 75° + 50° = 180°$ ✓.

3.

(a) $(10 - 2) \cdot 180° = 1440°$.

(b) Each exterior angle is $\frac{360°}{12} = 30°$, so each interior angle is $150°$. (Or $\frac{10 \cdot 180°}{12} = 150°$.)

(c) Exterior: $\frac{360°}{5} = 72°$. Interior: $180° - 72° = 108°$.

4.

(a) $\frac{1}{2} \cdot 10 \cdot 6 = 30$ cm².

(b) $8 \cdot 5 = 40$ cm². (The side of 6 cm is not needed; it is *not* the height.)

(c) $\frac{1}{2} \cdot (6 + 10) \cdot 4 = 32$ cm².

5.

Radius $r = 7$ cm. Circumference: $\pi d = 14\pi \approx 44.0$ cm. Area: $\pi r^2 = 49\pi \approx 153.9$ cm².

6.

(a) Volume: $4 \cdot 5 \cdot 6 = 120$ cm³. Surface area: three pairs of faces, $2(4 \cdot 5 + 4 \cdot 6 + 5 \cdot 6) = 2(20 + 24 + 30) = 148$ cm².

(b) Volume: $\pi \cdot 3^2 \cdot 10 = 90\pi \approx 282.7$ cm³ ≈ 0.28 litres (1 litre = 1000 cm³). Surface area: two circles $2 \cdot 9\pi = 18\pi$ and the curved side, a rectangle $2\pi \cdot 3$ by 10, so 60π. Total 78π cm² (≈ 245.0 cm²).

7.

	$A(-2,3)$	$B(4,1)$	$C(1,-2)$
(a) $(x,y)\mapsto(-x,y)$	$(2,3)$	$(-4,1)$	$(-1,-2)$
(b) $(x,y)\mapsto(x,-y)$	$(-2,-3)$	$(4,-1)$	$(1,2)$
(c) $(x,y)\mapsto(-y,x)$	$(-3,-2)$	$(-1,4)$	$(2,1)$
(d) $(x,y)\mapsto(x+3,y-1)$	$(1,2)$	$(7,0)$	$(4,-3)$

8.

Shape	Lines of symmetry	Order of rotation	Point symmetric?
square	4	4	yes
rectangle	2	2	yes
regular pentagon	5	5	no
parallelogram	0	2	yes
isosceles trapezium	1	1	no
H	2	2	yes
N	0	2	yes
A	1	1	no

A shape is point symmetric exactly when its order of rotational symmetry is even (then a half-turn is one of its symmetries).

9.

(a) $c^2 = 81 + 144 = 225$, $c = 15$.

(b) $b^2 = 289 - 64 = 225$, $b = 15$.

(c) $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. **Yes** (by the converse of Pythagoras).

(d) $5^2 + 6^2 = 61 \neq 64 = 8^2$. **No.** (Since $61 < 64$, the angle opposite the side 8 is larger than $90°$: the triangle is obtuse.)

School-exam level

10.

(a) $(180° - 36°) \div 2 = 72°$ each.

(b) $180° - 2 \cdot 50° = 80°$.

(c) The base angles are equal. If $100°$ were a base angle, the two base angles would add up to $200° > 180°$: impossible. So $100°$ is the top angle, and the base angles are $(180° - 100°) \div 2 = 40°$ each.

11.

(a) $10 \cdot 8 - 4 \cdot 3 = 80 - 12 = 68$ m².

(b) Go round the L-shape: the sides are 10, 8, and then, around the cut-out corner, pieces that together still go "across" and "up" by the same amounts as the missing corner of the rectangle. Explicitly, if the cut-out is in the top right corner: 10 (bottom), $8-3=5$ (right, lower part), 4 (inwards), 3 (up), $10-4=6$ (top), 8 (left). Total $10+5+4+3+6+8=36$ m².

The full rectangle has perimeter $2(10+8)=36$ m too. Cutting out a corner replaces the two outer sides of the corner (4 and 3) by two inner sides of the same lengths (3 and 4), so the perimeter doesn't change.

12.

(a) Height h: $h^2+1.4^2=5^2$, so $h^2=25-1.96=23.04$, $h=4.8$ m.

(b) Foot distance x with height 4: $x^2=25-16=9$, $x=3$ m. The foot moves $3-1.4=1.6$ m further from the wall.

13.

(a) $AB=\sqrt{(7-1)^2+(10-2)^2}=\sqrt{36+64}=\sqrt{100}=10$. Midpoint $\left(\frac{1+7}{2},\frac{2+10}{2}\right)=(4,6)$.

(b) AC is horizontal ($y=2$) and BC is vertical ($x=7$), so the angle at C is 90°. (Check with Pythagoras: $AC=6$, $BC=8$, $6^2+8^2=100=AB^2$ ✓.) Area $\frac{1}{2}\cdot 6\cdot 8=24$.

14.

One turn covers the circumference: $70\pi\approx 219.9$ cm. In 1 km $=100\,000$ cm: $100\,000\div 70\pi\approx 454.7$. So **454 full turns** (and part of a 455th).

15.

(a) $25\cdot 10\cdot 2=500$ m³ $=500\,000$ litres.

(b) $500\,000\div 20=25\,000$ s. $25\,000\div 3600\approx 6.94$ h: **6 hours 57 minutes** (exactly 6 h 56 min 40 s).

16.

(a) Draw $AB=6$ cm. Set the compass to 5 cm and draw an arc around A; set it to 4 cm and draw an arc around B. The arcs meet at C. Then $AC=5$ and $BC=4$.

(b) $2+3=5<6$. With the 6 cm side drawn, circles of radius 2 and 3 around its ends don't reach each other, so they don't meet. This is the **triangle inequality**: each side must be shorter than the other two together.

(c) Construction as in notes §9. P on the perpendicular bisector: reflecting in the bisector maps A to B and keeps P fixed, and reflection keeps lengths, so $PA=PB$. (Or: if M is the midpoint, triangles PMA and PMB have PM in common, $MA=MB$, and right angles at M, so by Pythagoras $PA^2=PM^2+MA^2=PM^2+MB^2=PB^2$.)

(d) **30°:** construct an equilateral triangle (60°) and bisect one of its angles. **45°:** construct a perpendicular (90°) and bisect the right angle.

17.

(a) **9.** From each of the 6 corners there are 3 diagonals (to the corners that are not itself and not its two neighbours), giving $6\cdot 3=18$, but each diagonal has been counted from both ends: $18\div 2=9$.

(b) Same argument: from each of the n corners, $n-3$ diagonals go out (all corners except itself and its two neighbours). That's $n(n-3)$, and each diagonal is counted twice, so $\frac{n(n-3)}{2}$.

(c) $\frac{10 \cdot 7}{2} = 35$.

(d) $\frac{n(n-3)}{2} = 20$ means $n(n-3) = 40 = 8 \cdot 5$. So $n = 8$: the **octagon**. (As n grows, $n(n-3)$ grows, so there is no other solution.)

18.

(a) Each exterior angle is $180° - 156° = 24°$, and they add up to $360°$, so $n = 360 \div 24 = 15$ sides.

(b) The exterior angle would be $80°$, and $360 \div 80 = 4.5$ is not a whole number. **No.**

19.

(a) $\pi \cdot 5^2 - \pi \cdot 3^2 = 25\pi - 9\pi = 16\pi \approx 50.3$ cm².

(b) The path is a ring with radii 4 and 5: $25\pi - 16\pi = 9\pi \approx 28.3$ m².

20.

(a) The minute hand turns $360° \div 60 = 6°$ per minute; at 40 minutes it is at $240°$ (measured clockwise from 12). The hour hand turns $30°$ per hour, $0.5°$ per minute; at 3:40 it is at $3 \cdot 30° + 40 \cdot 0.5° = 110°$. The angle between them is $240° - 110° = 130°$.

(b) m minutes after 3:00, the minute hand is at $6m$ degrees and the hour hand at $90 + 0.5m$. They coincide when $6m = 90 + 0.5m$, so $5.5m = 90$, $m = \frac{180}{11} = 16\frac{4}{11}$ minutes. $\frac{4}{11}$ of a minute is about 21.8 seconds. So at about **3:16:22**.

Challenge

21.

$MA = MB = MC$ (all radii). So triangles AMC and BMC are isosceles.

- In triangle AMC, the base angles at A and C are equal; call them x.
- In triangle BMC, the base angles at B and C are equal; call them y.

$\angle ACB$ is made of the two angles at C: $\angle ACB = x + y$.

The angles of triangle ABC are x (at A), y (at B) and $x + y$ (at C). So $x + y + (x + y) = 180°$, $2(x + y) = 180°$, and $\angle ACB = x + y = 90°$. □

(This is the first theorem of geometry with a known discoverer: Thales of Miletus, around 600 BC.)

22.

(a) At each corner the interior and exterior angle are on a straight line, so they add up to $180°$. Over all n corners: (interior sum) + (exterior sum) $= n \cdot 180°$. The interior sum is $(n-2) \cdot 180°$, so the exterior sum is $n \cdot 180° - (n-2) \cdot 180° = 2 \cdot 180° = 360°$. □

(b) Exterior angle $180° - 140° = 40°$; $360 \div 40 = 9$ sides.

23.

Where copies meet at a corner, the angles there must add up to exactly $360°$. All the angles are the interior angle $\alpha_n = 180° - \frac{360°}{n}$ of the regular n-gon, so $\frac{360°}{\alpha_n}$ must be a whole number k (the number of polygons meeting at a corner).

- $n = 3$: $\alpha = 60°$, $k = 6$ ✓.
- $n = 4$: $\alpha = 90°$, $k = 4$ ✓.
- $n = 5$: $\alpha = 108°$, $360 \div 108 = 3.33\ldots$ ✗.
- $n = 6$: $\alpha = 120°$, $k = 3$ ✓.
- $n \geq 7$: $\alpha_n = 180° - \frac{360°}{n}$ is more than $120°$ (since $\frac{360}{n} < 60$) and less than $180°$. Then $k = \frac{360°}{\alpha_n}$ lies strictly between 2 and 3, so it is not a whole number ✗.

So only $n = 3, 4, 6$ are possible, and these three really do tile the plane (think of graph paper, a honeycomb, and a triangular grid). □

24.

(a)

- **3 red faces:** the corner cubes: **8**.
- **2 red faces:** on an edge but not a corner. Each of the 12 edges has $3 - 2 = 1$ such cube: **12**.
- **1 red face:** in the middle of a face: **6** (one per face).
- **0 red faces:** the centre cube: **1**.

Check: $8 + 12 + 6 + 1 = 27$ ✓.

(b) For $n \times n \times n$:

- 3 faces: 8.
- 2 faces: $12(n-2)$.
- 1 face: $6(n-2)^2$.
- 0 faces: $(n-2)^3$ (the inner cube).

Check: with $m = n - 2$, $8 + 12m + 6m^2 + m^3 = (m+2)^3$, which you can verify by expanding $(m+2)^3 = (m+2)(m^2+4m+4) = m^3 + 6m^2 + 12m + 8$ ✓. So the four numbers add up to n^3.

25.

Put coordinates on it (this helps the picture): triangle 1 has its right angle at $(0, 0)$ and corners $(b, 0)$ and $(0, a)$. Triangle 2 has its right angle at $(0, a+b)$ and corners $(0, a)$ and $(a, a+b)$. The trapezium has corners $(0, 0)$, $(b, 0)$, $(a, a+b)$, $(0, a+b)$: its parallel sides (bottom and top) are b and a, and its height is $a + b$.

(a) At the point $(0, a)$ three angles lie along the vertical side: the angle of triangle 1 at that point, the angle of the third triangle, and the angle of triangle 2. The angles of triangles 1 and 2 at this point are the **two different acute angles** of the original right-angled triangle (in triangle 1 it is the angle opposite the leg b, in triangle 2 the angle opposite the leg a), so they add up to $90°$. The three angles together make $180°$, so the angle of the third triangle is $90°$. □

(b) **Trapezium formula:** $\frac{1}{2}(a+b)(a+b) = \frac{1}{2}(a+b)^2$. **Three triangles:** $\frac{1}{2}ab + \frac{1}{2}ab + \frac{1}{2}c^2$ (the third triangle is right-angled with both legs c).

So $\frac{1}{2}(a+b)^2 = ab + \frac{1}{2}c^2$. Multiply by 2: $a^2 + 2ab + b^2 = 2ab + c^2$, so $a^2 + b^2 = c^2$. □

26.

(a) Unfold the two faces the ant crosses into one flat rectangle; the shortest path is then a straight line, the diagonal of a rectangle. The rectangle has one side equal to one edge of the box and the other side equal to the sum of the other two edges. With edges 3, 4, 5 there are three ways:

Rectangle	Path
$(3+4)$ by 5	$\sqrt{49+25} = \sqrt{74} \approx 8.60$
$(3+5)$ by 4	$\sqrt{64+16} = \sqrt{80} \approx 8.94$
$(4+5)$ by 3	$\sqrt{81+9} = \sqrt{90} \approx 9.49$

The shortest is $\sqrt{74} \approx 8.6$ cm: the two short edges go together and the longest edge is separate.

(b) The space diagonal: $\sqrt{3^2+4^2+5^2} = \sqrt{50} \approx 7.07$ cm.

27.

(a) The smallest surrounding rectangle has $1 \le x \le 7$ and $1 \le y \le 8$: 6 by 7, area 42. Subtract the three right-angled corner triangles:

- between $A(1,1)$ and $B(7,3)$: legs 6 and 2, area 6;
- between $B(7,3)$ and $C(3,8)$: legs 4 and 5, area 10;
- between $C(3,8)$ and $A(1,1)$: legs 2 and 7, area 7.

Area of ABC: $42 - 6 - 10 - 7 = 19$.

(b) **Base and height:** base from $(0,0)$ to $(6,0)$ is 6, height (the y-coordinate of $(2,5)$) is 5: area $\frac{1}{2} \cdot 6 \cdot 5 = 15$. **Rectangle method:** rectangle 6 by 5 (area 30), minus triangles with legs 2 and 5 (area 5) and legs 4 and 5 (area 10): $30 - 15 = 15$ ✓.

28.

(a) Big square: $10^2 = 100$. The circle has diameter 10, so radius 5: area $25\pi \approx 78.5$. The small square has its diagonal equal to the diameter, 10. A square with side s has diagonal $s\sqrt{2}$ (Pythagoras: $s^2 + s^2 = d^2$), so $s^2 = \frac{d^2}{2} = 50$: area 50.

(Neat: the small square has exactly half the area of the big one. Rotate the small square by 45° and you can see it: its corners touch the midpoints of the big square's sides.)

(b) $\dfrac{25\pi}{100} = \dfrac{\pi}{4} \approx 0.785$, about 78.5%. It doesn't depend on the size: for a square of side s, the circle has area $\pi\left(\frac{s}{2}\right)^2 = \frac{\pi}{4}s^2$, always $\frac{\pi}{4}$ of the square.

29.

(a) Join the centre M to all six corners. This makes six triangles with two sides r and an angle of $\frac{360°}{6} = 60°$ at M. They are isosceles with top angle 60°, so the base angles are 60° too: they are equilateral. So each hexagon side is r, and the perimeter is $6r$.

The hexagon lies inside the circle, and each side is a straight line, shorter than the arc of the circle between the same two corners. So $6r < 2\pi r$, giving $\pi > 3$.

(b) The square around the circle has side $2r$ (the diameter), so perimeter $8r$. The circle lies inside the square, and its circumference is shorter than the square's perimeter: $2\pi r < 8r$,

so $\pi < 4$. (That a convex curve inside another is shorter is intuitively clear; it can be made rigorous later.)

(c) Join M to the six corners of the outer hexagon: again six equilateral triangles, but now the **height** of each (from M to the touching point in the middle of a side) is r. In an equilateral triangle with side s, the height splits it into two right-angled triangles with hypotenuse s and short leg $\frac{s}{2}$, so

$$r^2 = s^2 - \left(\tfrac{s}{2}\right)^2 = \tfrac{3}{4}s^2, \qquad s = \frac{2r}{\sqrt{3}}.$$

The outer hexagon's perimeter is $6s = \dfrac{12r}{\sqrt{3}} = 4\sqrt{3}\,r$. The circle is inside it, so $2\pi r < 4\sqrt{3}\,r$, giving $\pi < 2\sqrt{3} \approx 3.46$.

So $3 < \pi < 3.47$. Archimedes continued doubling the number of sides (12, 24, 48, 96) to get $3\frac{10}{71} < \pi < 3\frac{1}{7}$.